

Please stick the barcode label here.

(c) (i) At $t = 20$ s, what is the separation between cars A and B ?

(2 marks)

(ii) Deduce the time at which car B catches up with car A .

(2 marks)

(d) Both cars are of similar size and shape. It is known that the total resistive force experienced by each car is proportional to the square of its velocity. Determine the ratio of power output of the engine of car A to that of car B within the period $t = 20$ s to $t = 60$ s.

(2 marks)

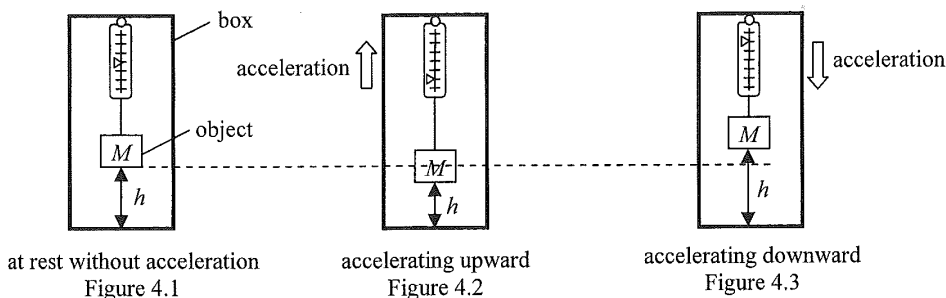
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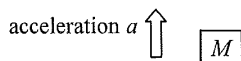
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4. Read the following description about **accelerometers** and answer the questions that follow.

An accelerometer is a device for measuring acceleration. The following example illustrates the principle of a simple accelerometer. An object of mass M is suspended by a spring balance inside a box. If the box is at rest without acceleration, the object is h above the bottom of the box (Figure 4.1). When the box accelerates upward, h decreases (Figure 4.2). Likewise, when the box accelerates downward, h increases (Figure 4.3). Since it is known that the tension of the spring balance is directly proportional to its extension, we can therefore determine the magnitude and direction of the box's acceleration by measuring h .



- (a) Draw a labelled free-body diagram in the space provided below showing the forces acting on the object when the box accelerates upward with acceleration a . Explain why h decreases in this case. (4 marks)



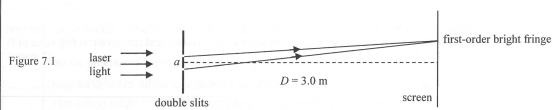
- (b) The scale of the spring balance is calibrated such that the pointer moves 1 cm for a change of 2 N of force. The weight of the object is 5 N. If h decreases by 0.5 cm compared to the situation in Figure 4.1, what is the reading of the balance? Hence find the magnitude of the corresponding acceleration of the box. (acceleration due to gravity $g = 9.81 \text{ m s}^{-2}$) (3 marks)

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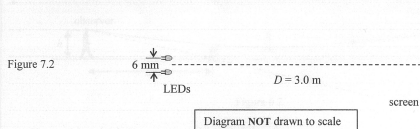
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7. * (a) A laser light beam of wavelength 650 nm is incident normally on a pair of slits separated by $a = 0.325\text{ mm}$. Interference pattern is observed on a screen at a distance $D = 3.0\text{ m}$ from the slits as shown in Figure 7.1. What is the separation between adjacent first- and second-order bright fringes? (2 marks)

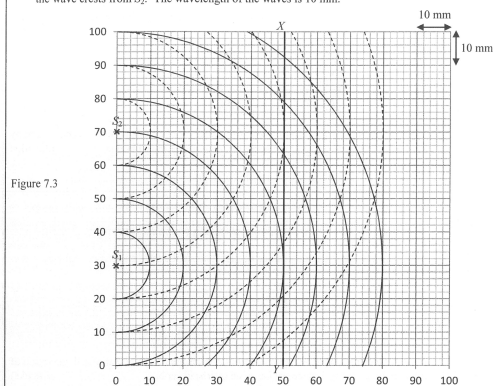


- (b) Figure 7.2 shows a set-up with two small LEDs separated by 6 mm and both LEDs emit light of wavelength 650 nm . State and explain what you would expect to see on the screen. (2 marks)



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Figure 7.3 shows circular water waves in a ripple tank. The two point sources S_1 and S_2 , separated by 40 mm , are driven by the same vibrator. The solid lines represent the wave crests from S_1 and the dotted lines represent the wave crests from S_2 . The wavelength of the waves is 10 mm .



- (c) Sketch on Figure 7.3 two lines to indicate all points P with path difference $PS_1 - PS_2$ equals to 10 mm (L_1) and 20 mm (L_2). State the kind of interference that occurs at these points P . (3 marks)

- (d) (i) If the interference pattern is observed along line XY at 50 mm from the sources as shown, measure the separation between adjacent first- and second-order maxima Δy . (1 mark)

separation $\Delta y =$ _____

- *(ii) However, using the calculation method in (a) would obtain 12.5 mm for this separation. Why does this calculated value differ with the measurement in (d)(i)? (2 marks)

Answers written in the margins will not be marked.

8. Electricity generated from power plants are transmitted at a high voltage through overhead cables in suburban areas.

(a) Each overhead cable consists of 40 strands of identical transmission lines bundled together.



a strand of transmission line of an overhead cable

(i) One single strand of transmission line has a cross-sectional area of $1.3 \times 10^{-5} \text{ m}^2$ and resistivity $2.6 \times 10^{-8} \Omega \text{ m}$. Find the resistance per km of a single strand of transmission line. (2 marks)

(ii) Explain why the resistance per km of an overhead cable is much smaller than that of a single strand of transmission line. Estimate the resistance per km of an overhead cable. (2 marks)

(iii) Hence, explain why a bird can stand with both feet on a high-voltage cable without getting an electric shock. (2 marks)



Answers written in the margins will not be marked.

* (b) Electrical power of 180 MW is transmitted at a voltage of 400 kV through an overhead cable.

(i) Calculate the current carried by the overhead cable. (2 marks)

(ii) Show that less than 0.1% of the electrical power is lost after transmitted through a total of 10 km of overhead cable. (2 marks)

(iii) As the voltage drop across this overhead cable is negligible, a voltage of 400 kV at the cable's end is stepped down by an ideal transformer with turns ratio 12 : 1.

(I) Find the secondary voltage from the transformer. (1 mark)

(II) State ONE factor leading to energy loss in a practical transformer and suggest the corresponding measure for improvement. (2 marks)

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(ii) Hence estimate the height of the building.

(2 marks)

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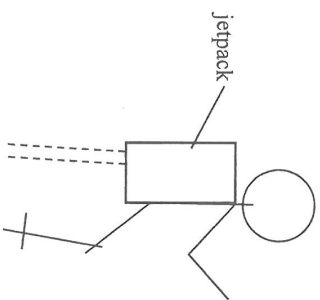
then ejected downwards.

- (a) Referring to Figure 4.1, water enters the U-shape hose inside the jetpack with a certain speed and is then ejected out vertically downwards. Use Newton's law(s) of motion to explain why a lifting force acting on the person is produced. (3 marks)

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Answers written in the margins will not be marked.

- (b) Draw and label all the forces acting on the person wearing the jetpack as a whole in the free-body diagram below. Neglect the pulling force due to the hose connected to the jetpack. (1 mark)



Answers written in the margins will not be marked.

2017-DSE
PHY
PAPER 1B

B

HONG KONG EXAMINATIONS AND ASSESSMENT AUTHORITY
HONG KONG DIPLOMA OF SECONDARY EDUCATION EXAMINATION 2017

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Candidate Number

PHYSICS PAPER 1

SECTION B : Question-Answer Book B

This paper must be answered in English

INSTRUCTIONS FOR SECTION B

- (1) After the announcement of the start of the examination, you should first write your Candidate Number in the space provided on Page 1 and stick barcode labels in the spaces provided on Pages 1, 3, 5, 7 and 9.
- (2) Refer to the general instructions on the cover of the Question Paper for Section A.
- (3) Answer **ALL** questions.
- (4) Write your answers in the spaces provided in this Question-Answer Book. Do not write in the margins. Answers written in the margins will not be marked.
- (5) Graph paper and supplementary answer sheets will be provided on request. Write your Candidate Number, mark the question number box and stick a barcode label on each sheet, and fasten them with string **INSIDE** this Question-Answer Book.
- (6) No extra time will be given to candidates for sticking on the barcode labels or filling in the question number boxes after the 'Time is up' announcement.

Question No.	Marks
1	7
2	5
3	4
4	10
5	8
6	10
7	11
8	12
9	10
10	7

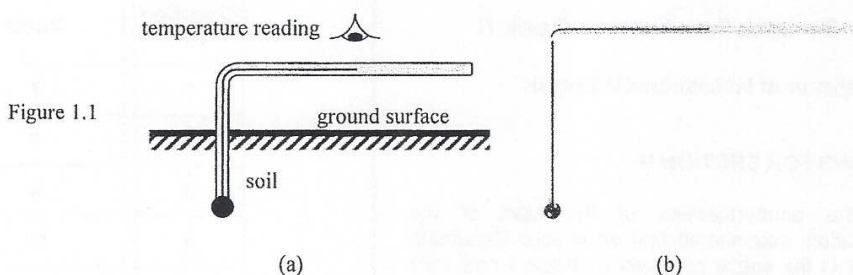
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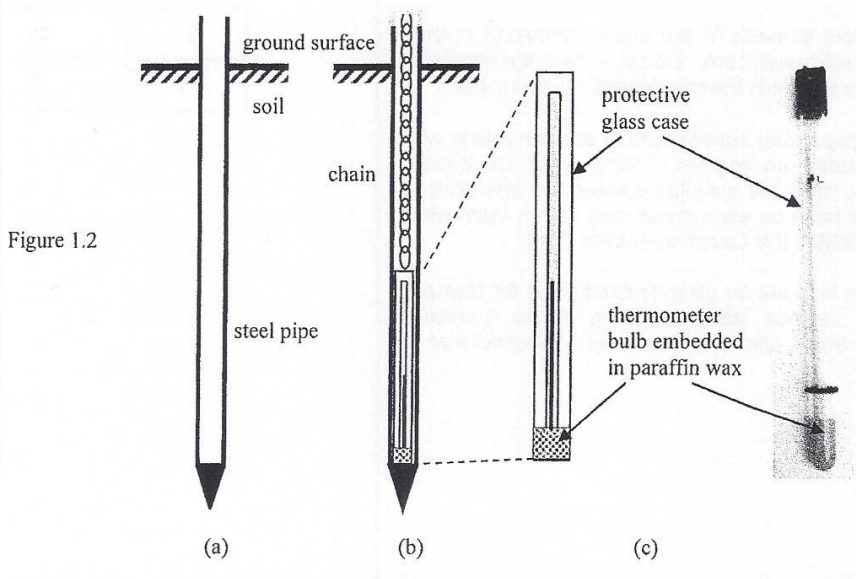
Section B: Answer **ALL** questions. Parts marked with * involve knowledge of the extension component. Write your answers in the spaces provided.

1. Read the following passage about **soil thermometer** and answer the questions that follow.

The temperature of soil changes with depth, and this information is important to farmers and scientists. To measure soil temperatures at depths close to the ground surface, the bulb of a thermometer is buried in the soil. The stem of the thermometer is bent 90° for easy reading. Figure 1.1a is a schematic diagram and Figure 1.1b shows a photo of a soil thermometer.



For depths greater than 30 cm, a steel pipe is driven into the soil (Figure 1.2a); and a liquid-in-glass thermometer with a protective glass case is lowered into the steel pipe (Figure 1.2b). The bulb of the thermometer is embedded in paraffin wax (Figure 1.2c). To read the temperature, the thermometer is lifted out of the steel pipe by pulling the chain.



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(1) In Figure 4.2, mark the positions of the projected bearing in the stroboscopic photo using circles (●). (2 marks)

(2) Given that the bearing is projected horizontally with an initial speed of 1 m s^{-1} , use the results of (a)(i) to calculate the speed of the projected bearing when the last image was taken. (3 marks)

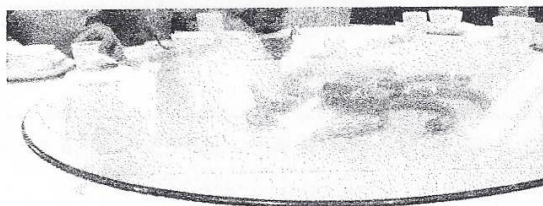
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(b) If a small ball is released from rest from the top of a cliff, the speed of the ball becomes constant after a period of time. By considering the forces acting on the ball and using Newton's laws of motion, explain why the speed of the ball becomes constant. (3 marks)

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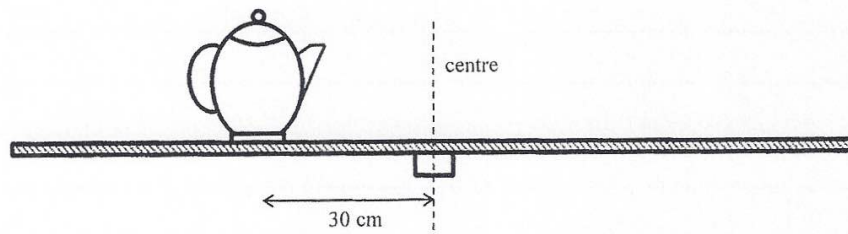
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*5. The photo shows a turntable commonly used in restaurants.



A teapot of mass 1 kg is put 30 cm from the centre of a horizontal turntable, Figure 5.1 shows the side view. When the turntable is rotating, the teapot remains at the same position on the turntable.

Figure 5.1



- (a) On Figure 5.1, draw and label all the forces acting on the teapot when the turntable is rotating. (2 marks)
- (b) Taking the teapot as a point mass, estimate the net force acting on the teapot when the turntable is rotating at a rate of 0.5 revolutions per second. (3 marks)

Answers written in the margins will not be marked.

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(ii) Hence find the speed of the sphere at point C .

(2 marks)

*(iii) If the horizontal distance between points C and E is 2.55 m, calculate the time of flight of the sphere before reaching point E .

(3 marks)

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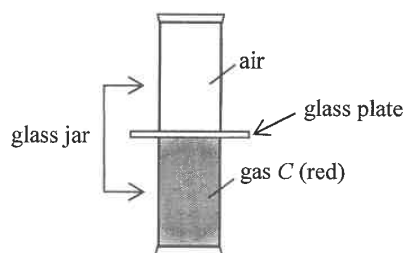
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- (II) Given that the root-mean-square speed ($c_{\text{r.m.s.}}$) of the molecules of gas A is 600 m s^{-1} , estimate $c_{\text{r.m.s.}}$ of the molecules of gas B . (2 marks)

- (b) Figure 2.3 shows two glass jars containing air and gas C respectively and they are separated by a glass plate. Both are at the same pressure and temperature. Gas C is red in colour.

Figure 2.3



After removing the glass plate, it takes a few minutes for gas C to diffuse several centimetres into the upper jar despite its molecules having a root-mean-square speed of 200 m s^{-1} . Explain this observation. (2 marks)

Answers written in the margins will not be marked.

3. Figure 3.1 shows a quadcopter with four propellers.



Figure 3.1

When the four propellers are operating to propel air streams vertically downwards, the quadcopter can hover in the air in a fixed position. ($g = 9.81 \text{ m s}^{-2}$)

- (a) Referring to Newton's laws of motion, explain why the quadcopter can hover in the air. (2 marks)

Given: mass of the quadcopter = 1.38 kg
total area swept by the **four** propellers = 0.284 m^2
density of air = 1.20 kg m^{-3}

- (b) Suppose the speed of the air streams produced is v .

- (i) Considering the total volume of air being propelled downwards in 1 second, express the mass of air m_a propelled by the quadcopter per second in terms of v . (2 marks)

- (ii) Hence, find the speed v that enables the quadcopter to hover. (2 marks)

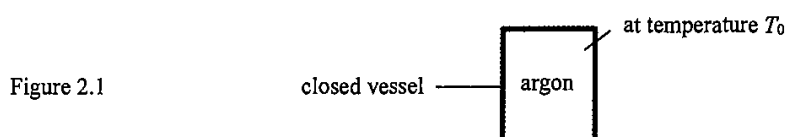
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- *2. Figure 2.1 shows a closed vessel of a fixed volume containing argon, a monatomic gas, maintained at temperature T_0 . The root-mean-square speed ($c_{\text{r.m.s.}}$) of the argon atoms inside is 500 m s^{-1} .



- (a) (i) Calculate the average kinetic energy of the argon atoms, E_K . Given that the mass of an argon atom is $6.63 \times 10^{-26} \text{ kg}$. (2 marks)

Answers written in the margins will not be marked.

- (ii) Hence, or otherwise, find T_0 (in K). (2 marks)

Answers written in the margins will not be marked.

- (b) If some of the argon has been leaked from the vessel while the temperature remains at T_0 , state and explain the change, if any, in the root-mean-square speed of the argon atoms inside the vessel. (2 marks)

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(a) Briefly describe the motion of the object from $t = 0$ s to $t = 4$ s.

(2 marks)

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(b) Find the position of the object at $t = 10$ s.

(2 marks)

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(c) Find the magnitude of the net force acting on the object at $t = 4$ s. The mass of the object is 0.4 kg.

(2 marks)

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